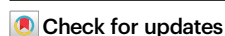


Total-area world-record efficiency of 27.03% for 350.0 cm² commercial-sized single-junction silicon solar cells

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Performance improvement is the cornerstone to facilitate the healthy and sustainable development of photovoltaic industry. Meanwhile, the aesthetics of solar panels becomes growingly concerned with the continuously improved requirements from customers. Accordingly, developing the modules having both a higher power conversion efficiency (PCE) and better aesthetic appearance is increasingly important. The structural advantage of back contact (BC) silicon solar cells, having a grid-line-free front surface, endows them with an exceptionally aesthetic appearance and the highest theoretical PCE among single-junction silicon solar cells. Fully utilizing these structural features is crucial for achieving high performance and gaining an insight into their industrial potential. Here, a facile double-sided light management strategy, incorporating hierarchical micro/submicrotextured pyramids on the sunny side and nanostructured polished surface in the rear gap region to reduce optical losses and improve appearance uniformity, has been developed on tunnel oxide passivated back contact (TBC) solar cells, to create a record total-area PCE of 27.03% for 350.0 cm² commercial-sized single-junction silicon solar cells. In addition, the low bifaciality factor that is the main short slab for BC technology is overcome by our TBC devices with the bifaciality factor of > 80%.

Silicon solar cells typically feature both top and rear contacts^{1–3}. The electrode grid on the front side blocks out incoming sunlight, reducing the available solar energy accordingly. The back contact (BC) device configuration has all electrodes placed on the rear side, offering the potential for a higher power conversion efficiency (PCE) and better aesthetic appearance, and thus has been drawing widespread interest. In 1975, Lammert and Schwartz at Purdue University first proposed the concept of interdigitated back contact (IBC) crystal silicon (c-Si) solar cells and managed to fabricate an IBC cell with the PCE exceeding 11%

by 1977^{4,5}. Subsequently, Swanson from Stanford University founded SunPower Corporation in 1985, marking the first commercial attempt at IBC technology. In 1993, SunPower provided customers with IBC cells having an average PCE exceeding 21%⁶, but the heavy reliance on the expensive photolithography technologies from the semiconductor industry made its first-generation IBC products less competitive in the mass market. Over 30 years of progress, with the exploitation of new process technologies such as low-cost laser patterning^{7,8}, advanced passivation contacts¹, and low-temperature metallization^{9,10}, the PCE of

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hybrid BC devices with rear heterojunction *p*-type contacts and tunnel oxide passivated contact (TOPCon) *n*-type contacts has reached 27.4% for a 165.72 cm² half-cut cell¹¹, and the PCE of a commercial-sized heterojunction BC cell has reached 26.93%⁸. However, despite the PCE of BC cells has far surpassed those of several mainstream cells in the past and at present such as aluminum back surface field (Al-BSF)^{12,13}, passivated emitter and rear cell^{14,15}, and TOPCon cells^{16,17}, high equipment investment and manufacturing costs, especially for heterojunction-related BC cells, as well as low bifaciality factors still limit the industrial scaling of BC technology.

Tunnel oxide passivated back contact (TBC) cells, which stem from the market-dominant TOPCon cells^{18,19}, have good compatibility with the production lines of TOPCon cells. The investment and renovation costs for the production lines of TBC cells are substantially lower than that of the BC cells based on heterojunction technology, making TBC technology the preferred choice for mass production. The PCE of TBC cells adopting passivated contact technology has broken through 25% in 2016 and 26% in 2018^{20–22}. Given the theoretical efficiency limit of 29.2%^{23,24}, it is of evidence that the structural advantages of TBC cells have yet to be harnessed. At present, a trend towards cost reduction in silicon solar cells has favored thinner silicon wafers^{25,26}, necessitating full absorption of incident light within a constrained wafer thickness, which poses higher requirements for the design of light-trapping structures within the cells. However, some effective silicon texturing techniques, such as moth-eye patterns and black silicon^{27–30}, bring severe electrical recombination losses along with optical gains, thus requiring a careful assessment of the side effects of the micro/nanostructures generated by these light-trapping strategies. Moreover, BC cells are normally criticized for their low bifaciality factors due to the materials/structures on the rear side, such as the heavily doped materials and metal grid lines, that may have parasitic light absorption and/or shade light³¹. The low bifaciality factor is a major reason for cautious stance of the market regarding BC technology.

In response to these concerns, we report a facile but efficient low-damage bifacial light management technique for TBC cells in this work. It mainly involves development of shallow step-like sub-microstructures on the front pyramid surfaces, which further reduce silicon wafer reflectance while maintaining low surface recombination properties. Concurrently, ball-crown-shaped nanostructures are formed in the polished gap regions on the rear side, which increase the internal reflection probability of rays and meanwhile reduce the surface recombination of carriers. Ultimately, a world-record PCE of 27.03% for 350.0 cm² commercial-sized single-junction silicon solar cells was created based on the TBC device configuration. Moreover, the bifaciality factor of our TBC devices is over 80%, clearing hurdles for the large-scale application of BC technology.

Results

350.0 cm² total-area TBC solar cells with 27.03% efficiency

TBC solar cells have an electrode-free front and the grooved gap regions on the rear side, providing ample space for light-trapping structure design. Such cells do not require consideration of front-side electrode contact issues, making it ideal for developing micro-nano structures to reduce optical losses on their sunny side. The rear side of a TBC cell features the *p*-type regions, *n*-type regions and gap regions. The contact in the *p*-type regions is typically comprised of an ultrathin SiO_x and *p*-type polycrystalline silicon (poly-Si), while the *n*-type contact consists of a tunneling SiO_x and *n*-type poly-Si, and the gap region is generally only composed of the silicon surface passivated by the Al₂O₃/SiN_x stack. Due to the difference in surface stresses and crystal orientations, the quality of the SiO_x film varies between polished and textured surfaces. The TOPCon structure exhibits significantly better passivation on a planar surface^{32–35}, which is verified in Supplementary Table 1. Thus, *p*- and *n*-type regions of highly efficient TBC cells are

commonly located on the polished surfaces, limiting the adjustability of the surface structure in these regions. However, for the gap regions, the silicon surface is directly passivated by the Al₂O₃/SiN_x stack, where the passivation effect is relatively less affected by the surface morphology, offering more choices in surface structure design.

Utilizing the structural features of TBC cells, this work designs a facile low-damage bifacial light-trapping structure for efficient light management. The fabricated cells, as shown in Fig. 1a, have a hierarchical micro/submicro pyramid-textured surface on the front side, and a planar surface with ball-crown-shaped nanostructures (hereafter called as nanostructured polished surface) in the gap regions. **With these light management structures, we achieved a certified PCE of 27.03% on a TBC solar cell (*p*-type region: 300 μm, *n*-type region: 200 μm, pitch: 650 μm, wafer thickness: 170 μm, low-temperature metallization) with an area of 350.0 cm² (Fig. 1b and Supplementary Fig. 1), which is also the highest total-area efficiency reported to date for single-junction silicon solar cells¹¹.** The short-circuit current density (*J*_{sc}) of the cell reaches 42.32 mA cm^{−2}, evidently higher than that of the high-efficiency BC cells reported recently (see Supplementary Table 2)^{8,10,20,21}, demonstrating the effectiveness of the adopted light management strategy in this work. As exhibited in Supplementary Fig. 2, since the efficiency of single c-Si solar cells exceeded 26%, their efficiency progress has stagnated, and the efficiency gap between them and silicon heterojunction cells has continued to widen¹¹. Fortunately, the TBC cells we fabricated here have achieved a breakthrough in efficiency, indicating that there is still much room to be explored in light-trapping structures on single c-Si cells. In this study, to explore suitable light management structures, several different texturing schemes were introduced and the corresponding device performance for TBC cells was compared. As described in Fig. 1c, the cells prepared by one-step etching (wet chemical etching 1) have smooth pyramid surfaces on both the front side and the rear gap regions, corresponding to a conventional microtextured structure (CMS). With the step of wet chemical etching 3, the conventional microtextured pyramids can be further modified to form a hierarchical micro/submicrotextured structure (HMS). Differentiated texturization of the front side and rear gap regions of the cells—initially forming microtextured pyramids on the front side through wet chemical etching 2 while forming a polished surface in the rear gap regions, followed by establishing an HMS surface on the front side and ball-crown-shaped nanostructures on the polished surface of the rear gap regions through wet chemical etching 3—yields a light-trapping structure termed the synergistic structure of HMS and nanostructured polished surface (NPS), i.e., HMNS, which is employed in our most efficient cells.

The cells with the aforementioned light management structures were fabricated on our mass production line (*p*-type region: 450 μm, *n*-type region: 330 μm, pitch: 920 μm, wafer thickness: 130 μm) and their photovoltaic (PV) parameters were statistically analyzed. Firstly, the concentration of the alkali etching agent in the modification process (i.e., wet chemical etching 3) was optimized to determine the appropriate HMS surface (see Supplementary Fig. 3). At the NaOH concentration of 3 wt%, the HMS-based solar cell achieved the highest average PCE. Moreover, it can be observed that as the NaOH concentration increases to 3 wt%, the *J*_{sc} improves gradually, while the open-circuit voltage (*V*_{oc}) and fill factor (FF) have no evident change, which will be explained in the following discussion. Next, under the optimum alkali concentration for the modification process, HMS- and HMNS-based cells were prepared for performance comparison with the CMS-based cells. As shown in Fig. 1d, the average PCE increases from 26.25% for the CMS-based cells to 26.41% for the HMS-based cells and 26.49% for the HMNS-based cells. The efficiency improvement for the HMS- and HMNS-based cells can be mainly attributed to the increased *J*_{sc} (Fig. 1e), which will be proven subsequently to benefit from the enhanced light trapping ability of the cells. Notably, the

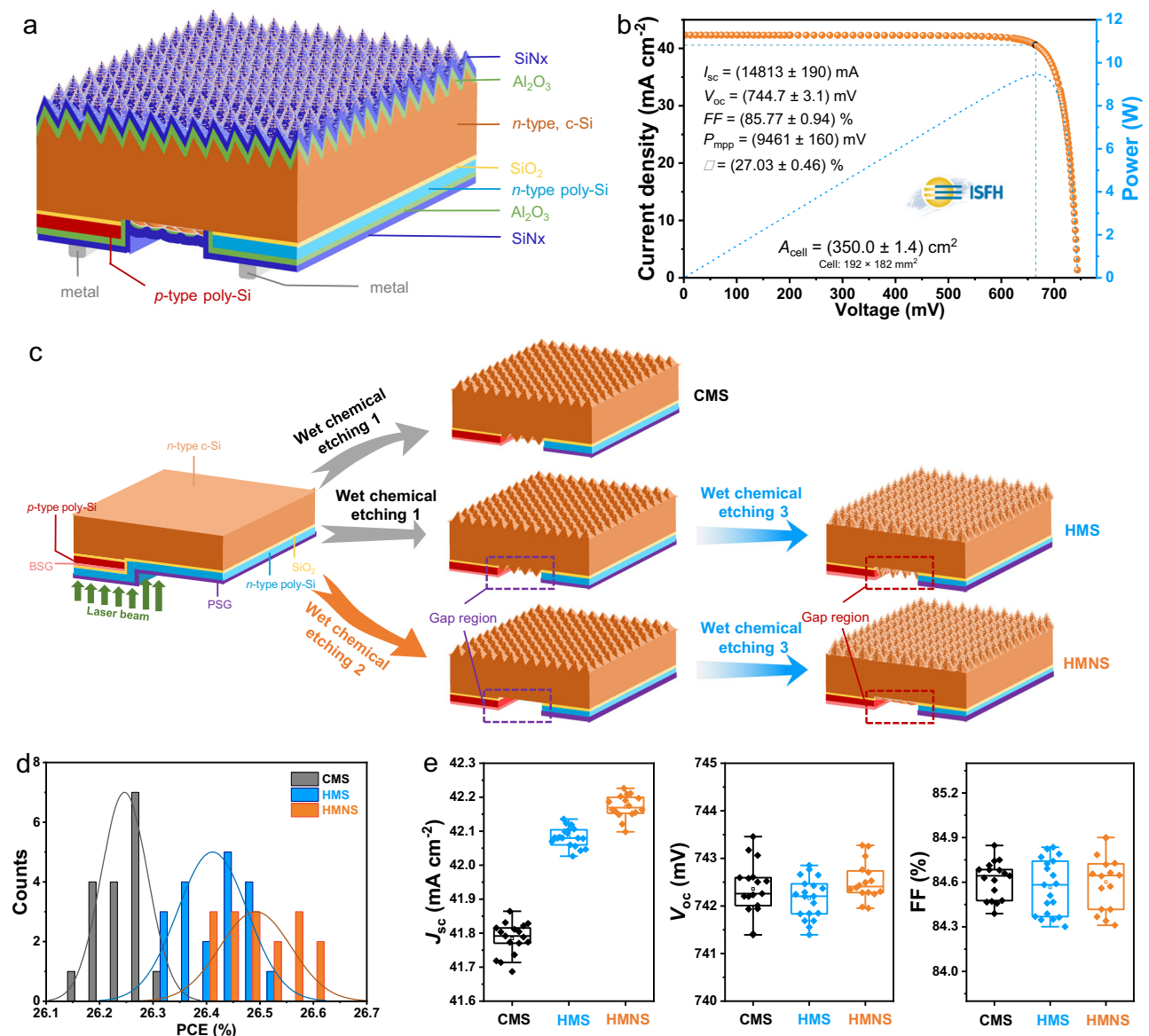


Fig. 1 | TBC solar cells in this study. **a** Diagram of the device configuration. **b** Current-voltage (*I*-*V*) and power-voltage (*P*-*V*) curves of the 350.0 cm² cell (*p*-type region: 300 μm, *n*-type region: 200 μm, pitch: 650 μm, wafer thickness: 170 μm) with the total-area world-record efficiency of 27.03% among commercial-sized single-junction silicon solar cells. **c** Schematic diagram of different texturing

processes and surface structures investigated here. **d** Efficiency distribution of our mass-produced TBC solar cells (*p*-type region: 450 μm, *n*-type region: 330 μm, pitch: 920 μm, wafer thickness: 130 μm) based on CMS, HMS, and HMNS. **e** Statistical J_{sc} (left), V_{oc} (middle), and FF (right).

HMNS-based cells show the highest average J_{sc} and a slight improvement in V_{oc} , indicating that a disparity in light trapping and passivation across the different light management structures. These results also demonstrate that exploiting the structural advantages of BC solar cells can contribute to further enhance their performance.

HMS surface for antireflection

The surface morphology plays a crucial role in reducing surface reflectance, largely determining the proportion of incident light that can refract into the silicon bulk^{36,37}. Light-trapping structures based on wave optics, such as moth-eye patterns and black silicon, which are equivalent to the media with a gradually varying refractive index, can reduce the reflectivity to <2%^{30,38,39}. However, the large specific surface area of such structures and the pollutants absorbed during their preparation process can lead to severe nonradiative recombination, so that the increase in current from the enhanced light trapping cannot

even compensate for the reduced V_{oc} and FF caused by carrier recombination.

To address this issue, we developed gently etched HMS pyramids for further reducing the front reflectance but not evidently sacrificing V_{oc} and FF of the cells. Conventional microtextured pyramids with a feature size of approximately 1 μm and relatively low heights (Fig. 2a) normally have smooth surfaces, still leading to a relatively higher reflection. A mild etching process (wet chemical etching 3) was further employed to reshape the surface of the conventional microtextured pyramids to form the pyramids with step-like submicrostructures, i.e., HMS pyramids (see Fig. 2b and Supplementary Fig. 4a, b). As depicted in Fig. 2c, the transition from CMS to HMS pyramids can be achieved by employing a class of polymer additives containing carbonyl groups. These carbonyl groups form weak or unstable interactions with hydroxyl groups on silicon surfaces, inducing dynamic adsorption-desorption processes of the polymers. This mechanism allows the

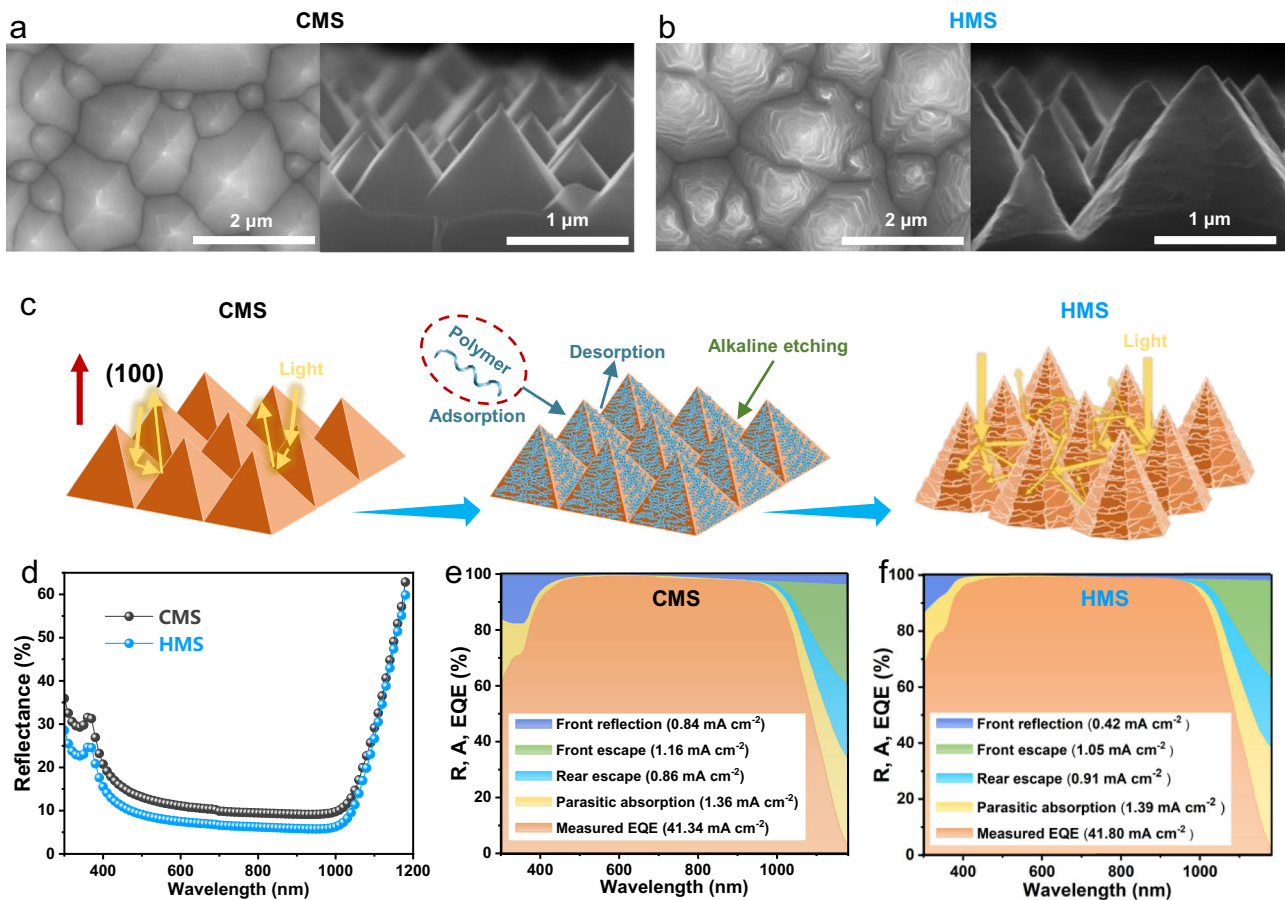


Fig. 2 | Reduced reflection with HMS surfaces. **a, b** Top-view (left) and cross-sectional (right) scanning electron microscope (SEM) images of the CMS (**a**) and HMS (**b**) pyramids. **c** Formation process of the HMS silicon surface. **d** Reflectance spectra of the CMS and HMS surfaces at normal incidence. **e, f** EQE spectra and

optical loss analysis of the mass-produced TBC solar cells (*p*-type region: 450 μm , *n*-type region: 330 μm , pitch: 920 μm , wafer thickness: 130 nm) based on CMS (**e**) and HMS (**f**).

alkaline solution to preferentially etch the uncovered silicon surfaces, ultimately enabling the fabrication of submicrostructures on the pyramid surfaces. As shown in Fig. 2d, the average reflectance (in the wavelength range from 300 to 1180 nm) of the HMS surface is about 13.47%, which is lower than that of 17.24% for the CMS surface. This reflection suppression is further transferred to the suppressed reflectance of the HMS surface after covering the $\text{Al}_2\text{O}_3/\text{SiN}_x$ antireflection/passivation stack, as compared with the CMS case (see Supplementary Fig. 4c). Reduced reflection can be ascribed to the better light-trapping mechanisms of the HMS textures. The base angle (relative to the substrate surface) increases from $\sim 52^\circ$ for the CMS pyramids to $\sim 60^\circ$ for the HMS ones (see Fig. 2a, b), which enhances the probability of multiple reflections accordingly. Moreover, the submicrostructures on the surface of the hierarchical pyramids, which are comparable in size to the incident light wavelengths, effectively couple incident light into the silicon substrate through enhanced light scattering^{40–43}, thereby further suppressing surface reflection.

The optical loss analysis documented in prior literature was adopted to more intuitively reflect the impact of the front surface structures on light trapping (Fig. 2e, f)⁴⁴. Clearly, the J_{sc} loss due to the front reflection decreases from 0.84 mA cm^{-2} for the CMS-based cell to 0.42 mA cm^{-2} for the HMS-based one, demonstrating the outstanding antireflection of the HMS surface. Furthermore, to distinguish the contributions of variations in the pyramid base angle and submicrostructures to the reduction of front reflection loss, we simulated the front reflection loss of smooth pyramid-textured surfaces as a function of the pyramid base angle using the PV Lighthouse wafer ray

tracer. As shown in Supplementary Fig. 4d, increasing the pyramid base angle from 52° to 60° reduces the front reflection loss by 0.24 mA cm^{-2} , which is substantially smaller than the difference (0.42 mA cm^{-2}) in the front reflection losses between the CMS- and HMS-based cells. This confirms that both the increased pyramid base angle and the submicrostructures on the HMS surface contribute synergistically to minimizing front reflection losses. Meanwhile, the J_{sc} losses because of the escaped solar irradiation from the front and rear sides, i.e., front escape and rear escape in the corresponding figures, of the HMS-based cell also decreases compared to that of the CMS-based cell, due to the enhanced light scattering resulted from the higher irregularity of the submicrostructures on the front surface to increase the optical path length and thus light absorption of the silicon substrate. Ultimately, the external quantum efficiency (EQE) results indicate that in comparison to the CMS-based cell with an integrated J_{sc} of 41.32 mA cm^{-2} , the HMS-based cell has a higher integrated J_{sc} of 41.80 mA cm^{-2} .

Furthermore, the cells with the HMS surface exhibit a darker and more uniform appearance, making them more convenient for sieving in the production line and more attractive for consumers. Supplementary Fig. 4e, f shows the front surfaces of the CMS- and HMS-based cells passivated by $\text{Al}_2\text{O}_3/\text{SiN}_x$ stacks when viewed at different angles. The CMS-based cell shows a significant variation in color between vertical and tilted viewing angles. In contrast, the HMS-based cell exhibits very limited difference in color when observed at different angles, all presenting a dark appearance. The difference in appearance can be attributed to the significantly distinct light-trapping

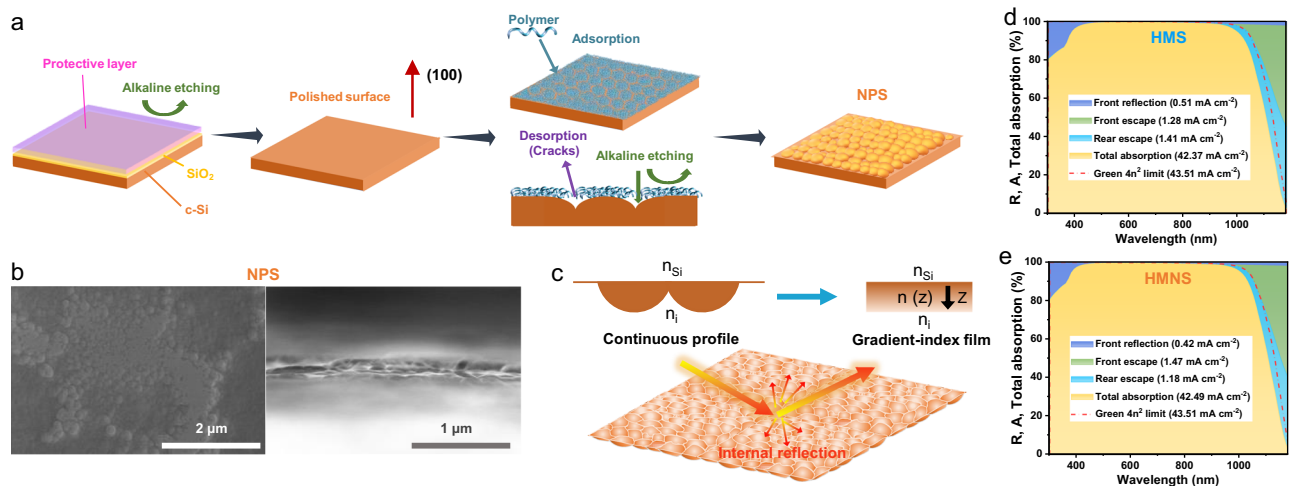


Fig. 3 | NPS surfaces. **a** Formation mechanism. **b** Top-view (left) and cross-sectional (right) SEM images. **c** Schematic diagram of the equivalent refractive index gradient. **d, e** Total absorption and optical loss analysis for the designed gap regions of HMS- (**d**) and HMNS-based (**e**) cells.

mechanisms between the CMS and HMS surfaces. The CMS surface primarily increases refraction into the silicon substrate through multiple reflections, yet its smooth crystal planes reflect light strongly in several directions. The HMS surface can effectively couple incident light into the silicon bulk through the enhanced light scattering that reduces the directionality of reflected light and thus elongates the optical path length, finally resulting in efficient light absorption and more consistent appearance for the cells accordingly.

Light trapping from synchronously formed nanostructured polished surface

Due to the relentless pursuit of the market on cost reduction and flexibility of silicon solar cells, the adoption of thinner silicon wafers is a major trend in the field^{25,26}. With the thinning of silicon wafers, the design of light-trapping structures in the cells becomes increasingly crucial. Rear polished surfaces in solar cells typically demonstrate superior light-trapping capabilities compared to pyramid-textured surfaces⁴⁵. Consequently, the gap regions are engineered to emulate the morphological characteristics of polished surfaces. Combining the characteristics of the TOPCon structure, we have achieved differentiated etching of the front side and rear gap regions of the TBC solar cells without any additional processing steps. After laser patterning, during the texturing process (wet chemical etching 2), a class of inorganic calcium salts is utilized as additives. In an alkaline environment, calcium ions from these inorganic salts interact with SiO₂, forming a protective layer on the SiO₂ film surface. This layer shields the underlying c-Si from alkaline etching, thereby enabling the formation of the CMS surface on the front side and a polished surface in the gap regions (Figs. 1c and 3a). The same mild etching process (wet chemical etching 3) as above was then adopted to treat the sample, forming the HMS surface on the front side (Supplementary Fig. 5a, b), and simultaneously forming ball-crown-shaped subwavelength structures on the polished surface, i.e., NPS, in the gap regions (as exhibited in Fig. 3a, b). The feature size of this subwavelength structure ranges from 30 to 250 nm, falling within the nano to submicron range, much smaller than the incident light. Accordingly, the light-trapping characteristics of the NPS surface can be explained using the effective media theory^{30,38}. The NPS surface works as an effective medium with a gradually varying refractive index, where the refractive index gradually decreases from the silicon bulk to air (Fig. 3c). Theoretically, due to the continuous refractive index gradient, light passing through the bulk silicon-air interface avoids abrupt refractive index changes. Additionally, the non-uniform structural features of the NPS surface may further enhance light scattering. Collectively, these effects endow the NPS

surface with superior light-trapping performance, which is experimentally verified in Supplementary Fig. 5c.

The enhanced light trapping of the HMNS-based cells is directly reflected by the improved J_{sc} and the corresponding EQE results (See Fig. 1e and Supplementary Fig. 5d). To more exactly verify the light-trapping advantages of the NPS surface, we prepared three wafer samples: Sample 1 having the CMS-based front and rear surfaces, Sample 2 having the HMS-based front and rear surfaces, and Sample 3 having the HMS front surface and NPS rear surface (i.e., the HMNS-based sample). As illustrated in Supplementary Fig. 5e and Fig. 3d, e, the optical loss analysis shows that the HMNS-based sample has the lowest escape losses of 2.65 mA cm⁻², including the front and rear escape losses, compared to the CMS-based sample (2.71 mA cm⁻²) and the HMS-based sample (2.69 mA cm⁻²), demonstrating the positive contribution of the NPS surface to light confinement. In addition, the front reflection loss also decreases from 0.93 mA cm⁻² for the CMS-based sample to 0.51 mA cm⁻² for the HMS-based sample and further to 0.42 mA cm⁻² for the HMNS-based sample, further confirming the effectiveness of the developed light management structures.

Passivation analysis for the designed surface structures

Due to the complexity of the highly efficient light-trapping structures, which typically accompany changes in morphology and surface area compared to conventional textured pyramids, it often leads to enhanced carrier recombination. Thus, surface passivation becomes a significant challenge for the valid design of light-trapping structures. To alleviate this problem, we have specifically optimized the Al₂O₃/SiN_x stacks, primarily the Al₂O₃ layer, on the HMS surfaces. Considering that the passivation ability of Al₂O₃ on c-Si surfaces mainly stems from field-effect passivation provided by the fixed negative charges at the interface and chemical passivation induced by hydrogen⁴⁶, the thickness of Al₂O₃ has been optimized. We passivated the HMS surfaces prepared with different NaOH concentrations. As displayed in Fig. 4a, the passivation difference of the Al₂O₃/SiN_x stacks on CMS and HMS can be negligible when the atomic layer deposition (ALD) circles of Al₂O₃ is more than 43 (~6 nm), suggesting that the nonradiative recombination of mildly etched hierarchical textured wafers can be effectively suppressed. According to the previous studies, when the ALD-deposited Al₂O₃ thickness is greater than 2 nm, the field-effect passivation remains essentially unchanged; therefore, the improved passivation performance can be mainly attributed to the enhanced hydrogen passivation⁴⁷.

Quasi-steady-state photoconductance lifetime measurement was performed on double-sided Al₂O₃/SiN_x-passivated samples to further

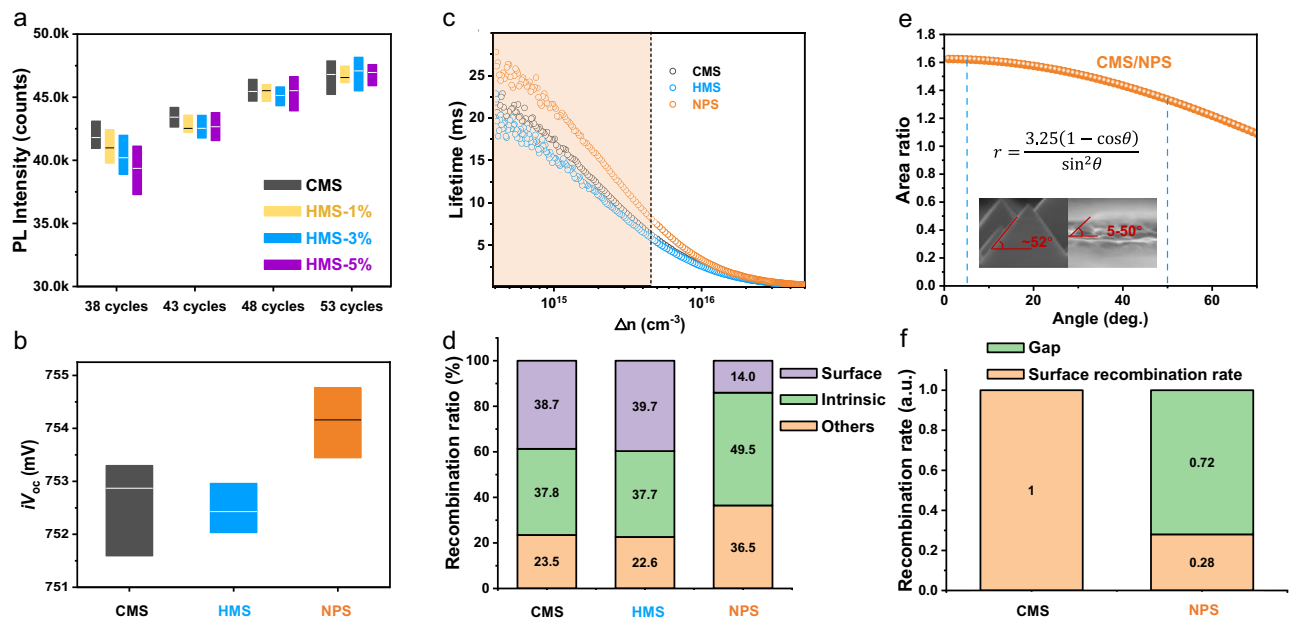


Fig. 4 | Passivation effect and recombination mechanism analysis. a Steady-state photoluminescence (PL) intensity of the silicon wafers with the CMS and HMS surfaces passivated by Al_2O_3/SiN_x stacks (HMS-1%, HMS-3%, and HMS-5% represent the HMS modified with NaOH concentrations of 1 wt%, 3 wt%, and 5 wt%, respectively, in the wet chemical etching 3). **b** iV_{oc} of the CMS-, HMS-, and NPS-based

samples. **c** Measured lifetime of the wafers with the CMS, HMS, and NPS surfaces. **d** Comparison of the surface, intrinsic, and SRH recombination ratios at the MPP injection level for different textured silicon wafers. **e** Surface area ratio of the CMS and NPS wafers with changing base angles. **f** Surface recombination comparison between the CMS and NPS wafers at the same injection level ($\sim 4.2 \times 10^{15} cm^{-3}$).

analyze the passivation effects and corresponding recombination mechanisms of the CMS, HMS, and NPS surfaces. We found that the recombination current density (J_0) of the CMS and HMS surfaces are similar (Supplementary Fig. 6a), and their implied V_{oc} (iV_{oc}) are basically consistent (Fig. 4b), thereby further demonstrating that the HMS surfaces we designed can be effectively passivated like the conventional pyramid textures, i.e., the CMS surface. And it can be observed that the NPS sample exhibits significantly lower J_0 and higher iV_{oc} , indicating that the NPS surface has considerable advantages in reducing carrier recombination. Figure 4c presents the lifetime profiles of these three samples within the mid-high injection range, a region typically associated with the location of the maximum power point (MPP) for high-efficiency TBC solar cells. The observed enhancement in lifetime for the NPS-based sample implies that the corresponding cells could have a higher FF, which will be confirmed in the next section of this study.

To distinguish the differences in recombination mechanisms on different structured surfaces, as presented in Supplementary Fig. 6b, the effective recombination lifetime (τ_{eff}) can be deconstructed into three constituents: intrinsic recombination lifetime ($\tau_{intrinsic}$), surface recombination lifetime ($\tau_{surface}$), and other recombination lifetime (τ_{others}). The $\tau_{intrinsic}$ encompasses contributions from two distinct processes, namely, radiative recombination and Auger recombination, and can be quantitatively calculated as reported in previous studies^{48,49}. The $\tau_{surface}$ can be derived from the previous methods^{25,50,51}, and then the τ_{others} , dominated by Schottky-Read-Hall (SRH) recombination of impurities and defects, can be fitted. To determine the MPP injection levels for the CMS-, HMS-, and HMNS-based cells, the corresponding precursors were first prepared (p -region: 450 μm , n -region: 330 μm , pitch: 920 μm ; wafer thickness: 130 μm). Subsequently, at the MPP injection level, the differences in recombination mechanisms on different structured surfaces were analyzed as shown in Fig. 4d. The proportion of surface recombination is similar between the CMS and HMS surfaces, and it is significantly lower for the NPS surface, once again illustrating that the designed light-trapping structures have low surface damage. It is noteworthy that although the surface area of the

CMS surface is only about 1.4–1.6 times that of the NPS surface (Fig. 4e), its surface recombination velocity reaches more than 3 times that of the NPS surface (Fig. 4f). In accordance with the previous reports, this phenomenon can be explained by differences in surface morphology and crystal orientation, with smooth surface structures having better passivation advantages⁵². The above results prove the tremendous potential of the HNMS textures in achieving high-performance silicon solar cells.

Simultaneous improvement of bifaciality and efficiency

Centralized PV systems dominate the main market in PV applications, and the rear power generation capability is a core indicator for assessing the suitability of solar cells for centralized power generation. For a long period, BC cells have been not favored for their low bifaciality factors, which make their application scenarios mainly confined to the distributed PVs. Therefore, enhancing the bifaciality factors of BC cells is essential to broaden their applications. The factors affecting the bifaciality of TBC cells can be concluded into several aspects according to their structural characteristics, mainly including the parasitic absorption from heavily doped poly-Si, the shading by the grid lines, and the surface morphology on the rear side³¹. The polished rear surface is the mainstream choice for high-efficiency silicon solar cells, with the electrode grid lines of appropriate widths and densities to facilitate charge transfer, making the optimization of the heavily doped poly-Si in p - and n -type regions a preferred direction to improve bifaciality. It can be speculated that the bifaciality factors of mass-produced TBC cells based on CMS, HMS, and HMNS vary with the widths of the p - and n -type regions according to experimental results and PV Lighthouse simulations. As indicated in Supplementary Fig. 7a, b and Fig. 5a, the bifaciality factors of the BC cells can be promoted to over 80% with the reduction of the p - and n -type region widths. This improvement is attributed to the significant decrease in parasitic absorption of the back-incident light caused by the p - and n -type regions as the width of the gap region increases. To check the simulation results, we fabricated 8 independent cells for both the HMS- and HMNS-based configurations with the p - and n -type region widths

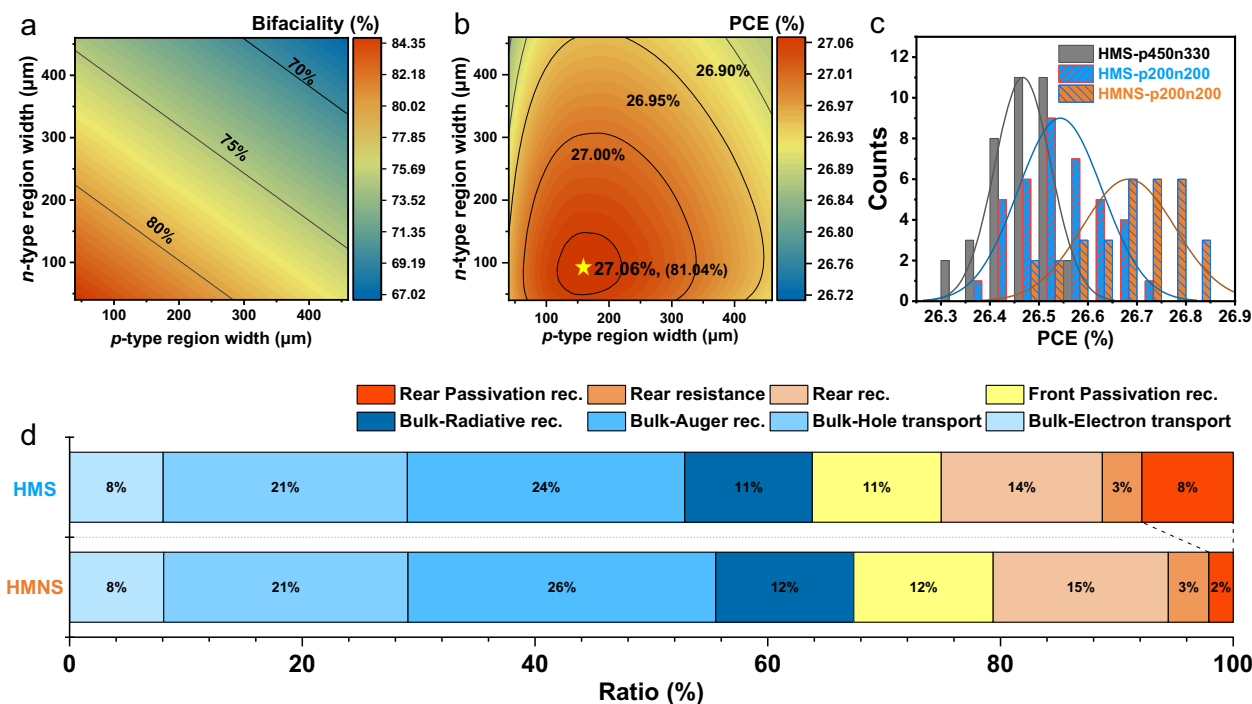


Fig. 5 | Simultaneous improvement in bifaciality and efficiency for the mass-produced TBC solar cells. **a** Bifaciality factor prediction of the HMNS-based devices. **b** Efficiency prediction of the HMNS-based devices. **c** Statistical efficiency distribution of the fabricated TBC devices (HMS-p450n330 is based on the HMS configuration with the corresponding widths of *p*-type and *n*-type regions of 450 and 330 μm; HMS-p200n200 has the same device configuration with HMS-

p450n330 but different widths of both *p*-type and *n*-type regions of 200 μm; HMNS-p200n200 is based on the HMNS configuration and has the widths of both *p*-type and *n*-type regions of 200 μm). **d** Side-to-side comparison of electrical loss components for the HMS- and HMNS-based TBC solar cells with the *p*-type region of 160 μm and *n*-type region of 90 μm. Here, the thickness of the relevant silicon wafers is 130 μm, and the pitch width of the relevant cells is fixed at 920 μm.

of 200 μm and the pitch width of 920 μm based on the 130-μm-thick wafers, and the corresponding average bifaciality factors are 81.33% and 77.94%, well consistent with the simulation values of 81.85% and 77.75% (see Supplementary Fig. 7c).

Referring to the experimental results of EQE, *I*-*V*, and optical loss analysis, we further corrected the J_{sc} of the Quokka3-simulated cells, thereby predicting the efficiency trend of our mass-produced TBC cells based on low-temperature metallization (Supplementary Fig. 8 and Fig. 5b). It is evident that suitable narrowing of the *p*- and *n*-type region widths is predicated to facilitate a higher efficiency. The efficiency of mass-produced TBC cells (*p*-type region: 450 μm, *n*-type region: 330 μm, pitch: 920 μm, wafer thickness: 130 nm) based on HMNS is anticipated to reach up to 27.06%, which is higher than 26.90% for the HMS-based cells, and 26.72% for the CMS-based cells. However, their corresponding bifaciality factors are essentially similar, each around 80%, remarkably higher than that of the reported BC cells. Notably, as the widths of the *p*- and *n*-type regions decrease, the increase of the predicted efficiency of the HMNS-based cells (−0.15%) is markedly higher than that of the CMS- and HMS-based cells (−0.05–0.07%), as meanwhile verified by the experimental results illustrated in Figs. 1d and 5c. Based on the prior discussion, the difference in efficiency gains should be primarily associated with the surface structures of the rear gap regions. When a CMS or HMS surface is employed in the gap regions, there would be the increased surface recombination and charge collection loss, leading to the dropped FF and thus less pronounced improvement in efficiency (Supplementary Fig. 9c and Fig. 5c). In contrast, when the rear gap regions adopt a nanostructured polished surface, benefiting from its excellent passivation effect and light confinement capability, the overall recombination losses and optical losses of the cells are significantly reduced as the gap region widens, resulting in a more substantial increase in efficiency. Furthermore, as depicted in Fig. 5d and Supplementary

Fig. 9d, the electrical loss analysis results reveal that the surface recombination loss within the gap regions of the HMNS-based cells is significantly lower than that of the HMS-based cells, explaining the higher efficiency boost for the HMNS-based cells relative to the CMS- and HMS-based cells. Therefore, by fully exploiting the structural features of TBC cells and designing rational surface light management structures, the improvement of both bifaciality and efficiency can be achieved.

Discussion

Through developing a facile light management strategy, the total-area world-record PCE of 27.03% for 350.0 cm² commercial-sized silicon solar cells was created based on the TBC configuration with the *p*-type region of 300 μm, *n*-type region of 200 μm, pitch width of 650 μm and 170-μm-thick phosphorus-doped Czochralski monocrystalline silicon (100) wafers. In this strategy, the front of the cells is designed as a low-damage micron/submicron composite structure prepared by wet-etching the conventional pyramid texture to form shallow step-like microstructures on the pyramid surface. This front surface effectively suppresses the front reflection of the cells, thereby improving their efficiencies and appearance uniformity. The surface of the rear gap region is designed to be a planar surface with ball-crown-shaped nanostructures, which has been proven to enhance the internal light-trapping and overall passivation effects of the cells. Moreover, the low bifaciality inherently associated with BC cells can be overcome by introducing the bifacial light management structure, and >80% bifaciality factor has been achieved by our TBC cells.

Methods

Fabrication of TBC solar cells

In this work, phosphorus-doped *n*-type Czochralski monocrystalline silicon (100) wafers of 350.0 cm² with a resistivity of 6–12 Ω·cm were

used for cell fabrication. The wafers were first cleaned and polished to 130 μm for subsequent deposition (Note: The wafers for certified cells were polished to 170 μm). A 1.7-nm-thick SiO_2 layer and a 300-nm-thick intrinsic amorphous silicon (i-a-Si) layer were deposited sequentially on the surface of the pretreated wafers by low-pressure chemical vapor deposition. BrCl_3 and O_2 were then used as the process gases to complete the deposition of boron-doped poly-Si and borosilicate glass (BSG) in a diffusion furnace. After laser crushing of BSG and wet removal of p -type poly-Si in the gap region and n -type region, a 1.5-nm-thick SiO_2 layer and a 200-nm-thick i-a-Si layer were grown on both sides of the samples. In a subsequent high temperature step under PCl_5 and O_2 atmosphere, phosphorus-doped poly-Si and phosphosilicate glass (PSG) were prepared on the surface of the sample. Subsequently, the PSG in the p -type region and gap region of samples was removed by laser. The front side of the samples was polished by HF/HNO_3 in water. For the CMS-based cells, the samples were then directly treated with NaOH/IPA mixed solution at a temperature of 68 $^\circ\text{C}$ for 6 min (i.e., wet chemical etching 1) to achieve the conventional pyramid texture on the front surface and gap region. For the HMS-based cells, the pyramids prepared by wet chemical etching 1 were further reconstructed at 70 $^\circ\text{C}$ for 2 min by NaOH solution with a polymer additive (i.e., wet chemical etching 3, additive concentration: 0.01–1 vol.%), which adsorbed onto the surface of crystalline silicon through weak forces. For the HMNS-based cells, the samples with the acid-polished front surface were immersed in the mixture solution of NaOH , IPA , and a type of inorganic calcium salts at a temperature of 68 $^\circ\text{C}$ for 6 min (i.e., wet chemical etching 2). Then, the front surface and rear gap region of the samples were reshaped through the wet chemical etching 3. After texturing, the samples were etched in HF solution to remove remaining BSG and PSG masks. Then, Al_2O_3 and SiN_x were sequentially deposited onto the surface of the samples using ALD and plasma-enhanced CVD techniques, respectively. Finally, a sliver paste was screen-printed onto the contacts and fired at 780 $^\circ\text{C}$ in air ventilation to form electrodes. Specially, a pulsed ultraviolet picosecond laser would be employed to peel off local $\text{Al}_2\text{O}_3/\text{SiN}_x$ stacks for contact opening when the low-temperature metallization route is adopted. For the low-temperature metallization, a sliver-coated copper paste was screen-printed onto the contacts and anneal at 250 $^\circ\text{C}$ in N_2 atmosphere. After fabrication, a high-intensity illuminated annealing was performed under 10–20 suns and 400 $^\circ\text{C}$ for 60 s on the cells using the light-emitting diode (LED) matrix (wavelength: 400–800 nm).

Samples for passivation analysis

All samples were fabricated using the wafers selected from the same position of the same silicon ingot as used in the TBC solar cells to ensure identical bulk properties such as resistivity and SRH recombination. A 1.5-nm-thick SiO_2 layer and a 200-nm-thick phosphorus-doped poly-Si were deposited onto the polished wafers to complete gettering process. Subsequently, the wafers with the double-sided CMS, HMS, and NPS surfaces were prepared using different laser etching and wet chemical etching processes. The $\text{Al}_2\text{O}_3/\text{SiN}_x$ stacks were then deposited symmetrically on the both sides of the textured wafers. Finally, a light soaking under 10–20 suns was carried out at 400 $^\circ\text{C}$ for 60 s using the LED matrix.

Characterization

The I - V characteristics of the solar cells were conducted on a Vision VS-6821S IV tester under the standard test condition (AM 1.5G: 100 mW cm^{-2} , 25 $^\circ\text{C}$). More than 15 cells per group were included for statistical analysis. The state-of-the-art TBC solar cell was certified by the Institute for Solar Energy Research in Hamelin (ISFH), Germany. Without any mask, all cells were measured on a total area of 350.0 cm^2 in ambient air. The total area of 350.0 cm^2 was defined by self-testing and ISFH certification. The surface morphology and cross-sectional images of the textures were observed by a SEM (Apreo S, Thermo

Scientific), The EQE, reflectance, and transmittance spectra were recorded with a quantum efficiency measurement system (PVE300-IVT210) from Industrial Vision Technology (S) Pte Ltd, and the light intensity was calibrated with standard single-crystal silicon and germanium reference solar cells. The thicknesses of the films involved were evaluated from a JAW T-solar Ellipsometer. The steady-state PL intensity of the samples was obtained from a PL detection system (LIS-R3, BT Imaging Pty Ltd). The τ_{eff} , J_0 , and iV_{oc} were acquired from a silicon wafer lifetime tester (WCT-120TS, Sinton Instruments) in a transient mode.

Device simulations

The Quokka3 simulations were performed to predict optimal device performance and guide cell design. The input parameters primarily come from the characterization results. The series resistance of the external circuit and initial optical file were calculated through PV Lighthouse. The simulated current densities have been adjusted based on the characterization results of EQE, I - V , reflectance, etc. The resistivity, thickness, and bulk lifetime of the silicon wafers were set to 8 $\Omega\text{-cm}$, 130 μm , and 40 ms, respectively. And the J_0 of the CMS, HMS, and NPS surfaces were set to be 0.68, 0.73, and 0.18 fA cm^{-2} , respectively.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated or analyzed during this study are included in the published paper and its Supplementary Information and Source Data files. Source data are provided with this paper.

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Author contributions

H.T., S.T., J.S.L., and Z.L. together led the project, with contributions from all authors. H.T. and S.T. conceived the idea. C.D., H.Z., J.H., J.C., X.X., C.C., Y.C., H.S., L.Y., and J.Y.L. fabricated the TBC solar cell. H.T. and S.T. designed the characterization experiments. Y.Z., Y.H., and H.L. conducted the characterization and measurement. S.T., Y.Z., Y.H., Y.L., J.K., F.S., X.X., B.F., X.J., and D.H. analyzed the data and explored the mechanisms. S.T., Y.Z., and X.X. performed device simulations. H.T. and S.T. co-wrote the manuscript. J.S.L. and Z.L. supervised the study.

Competing interests

The authors declare no competing interests.

Additional information

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